

Multinomial Distribution

While Bernoulli events can only have one of two possible outcomes (success or failure), this distribution models scenarios where we have multiple possible outcomes, hence the name *multinomial*.

Each experiment has a finite m number of possible outcomes.

$$X = (X_1, X_2, \dots, X_m)$$

$$p = (p_1, p_2, \dots, p_m)$$

We then have the PDF, modelling the number of occurrences of each possible outcome in n independent trials:

$$f(x, p, n) = \frac{n!}{\prod_{i=1}^m x_i!} \prod_{i=1}^m p_i^{x_i}$$

$$\sum_{i=1}^m x_i = n, \sum_{i=1}^m p_i = 1$$

Negative Multinomial Distribution

This model generalizes the **Pascal Distribution**, in which the multinomial experiment is repeated until k_0 occurrences of the result X_0 are observed.

$$X = (X_1, X_2, \dots, X_m)$$

$$p = (p_1, p_2, \dots, p_m)$$

$$p_0 = 1 - \sum_{i=1}^m p_i$$

$$E[X] = \frac{k_0}{p_0} p$$

$$Var[X] = \frac{k_0}{p_0^2} + \frac{k_0}{p_0} diag(p)$$